Phan Anh VU (♂)

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AI and chip design research engineer

Experience

PhD Researcher, *CEA (Commissariat à l'Énergie Atomique et alternative)*

2023/03 - 2026/07 Grenoble, France

- Develop novel methods for efficient uncertainty quantification in deep learning, with a focus on hardware-software co-design. Implement the proposed method in Jax, using Vision Transformer and MLPMixer architecture, and benchmark against existing methods.
- Implement the proposed algorithm in hardware. Design a novel analog in-memory computing circuit using memristor with Cadence Virtuoso. Simulate the circuit with a CMOS technology process using Eldo.

Intern Researcher, *ENS Paris Saclay SATIE Lab*

2022/02 - 2022/08 Paris, France

Study the prospects of using probabilistic and physics-informed machine learning for predictive maintenance of power electronics with time series data. Benchmark PyTorch implementations of various methods for uncertainty quantification in deep learning.

Intern Researcher, *Université Paris Saclay LISN Lab*

2021/04 - 2021/09 Paris, France

Benchmark PyTorch implementations of various methods for few-shot learning and meta-learning on datasets across different domains.

Software Engineer, *Cegid Talentsoft*

2017/11 - 2019/12 Paris, France

- For a talent management web application with estimated 9 million users and 2000 institutional clients, diagnose and resolve complex critical bugs, of various aspects such as front-end (JavaScript and TypeScript), back-end (C#, .NET), database (SQLServer).
- Implement new features, developer tools.

Skills

Machine Learning & Data Science: Python, Jax, PyTorch, Keras, TensorFlow, numpy, pandas, scikit-learn, matplotlib, seaborn, Julia

Electronic Design Automation (EDA): Cadence Virtuoso, Eldo, VHDL

Software Development: C#, .NET ASP MVC, JavaScript, TypeScript, Git, SQL, Redis

Publications

Adaptive Low-Rank Ensemble: estimation of ensemble performance as a function of the size, and reduce memory size by 3 up to 6 with equal performance as traditional ensemble (patent pending, preprint available on request).

Compressed vector-matrix multiplication for Memristor-based ensemble neural networks (patented), *IEEE International Conference on Rebooting Computing 2024*: an analog in-memory computing circuit with memristor to implement vector element-wise multiplication and low-rank approximation of ensemble of neural networks (Blog page).

Probabilistic and Physics-Informed Machine Learning for Predictive Maintenance with Time Series Data, *EuroSimE 2023*: (Code repository)

Meta-Album: Multi-domain Meta-Dataset for Few-Shot Image Classification, *NeurIPS 2022 Datasets and Benchmarks*: (Website)

Education

PhD, Machine Learning & Electronics, *Université Grenoble Alpes*

2023 - 2026 Grenoble, France

Master, Computer Science & Artificial Intelligence, *Université Paris Saclay*

2020 - 2022 Paris, France

Languages

English, Français, Tiếng Việt ●●●

中文, Español, Deutsch ●○○